

TECHNICAL ENGINEERING REPORT

Valorization of Biomass Through Anaerobic Digestion and Hydrothermal Carbonization: Integrated Process Design

Course	MEG 315 — Applied Thermodynamics
Institution	University of Lagos (UNILAG)
Department	Mechanical Engineering
Group	Group 28
Date	February 2026

Based on: Potrč, S., Petrovič, A., Egieya, J. M., & Čuček, L. (2025). Valorization of biomass through anaerobic digestion and hydrothermal carbonization: Integrated process flowsheet and supply chain network optimization. *Energies*, 18(2), 334. <https://doi.org/10.3390/en18020334>

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1. Abstract

This report examines the thermodynamic design and process integration of an Anaerobic Digestion–Hydrothermal Carbonization (AD-HTC) biorefinery system, contextualised within the MEG 315 Applied Thermodynamics curriculum at the University of Lagos. Building on the supply chain and process optimisation methodology of Potrč et al. (2025), this report develops the thermodynamic analysis of the system with focus on gas power cycle modelling, vapour power cycle analysis, and heat integration via Pinch Analysis.

The integrated system converts wet biomass — including locally relevant feedstocks such as cassava peel, sugarcane bagasse, and palm empty fruit bunch — through anaerobic digestion to produce biogas and digestate. Biogas fuels a Combined Heat and Power (CHP) unit modelled as an Otto cycle internal combustion engine. Digestate is valorised through hydrothermal carbonization at temperatures of 180–260°C to yield hydrochar with a higher heating value of up to 24,000 kJ/kg. CHP exhaust waste heat is cascaded to satisfy both the AD mesophilic heating requirement (37°C) and the HTC reactor thermal input, with analysis of the resulting heat surplus or deficit via Pinch methodology. The steam bottoming cycle is analysed as an ideal Rankine cycle at a fixed boiler pressure of 6.0 MPa (60 bar), with state-point calculations on the Mollier h-s diagram.

This analysis demonstrates that thermodynamic optimisation at the system level — integrating cycle efficiency, heat recovery, and mass balance — is the defining engineering challenge of the AD-HTC biorefinery, and cannot be resolved by optimising individual unit operations in isolation.

2. Introduction

The global transition toward circular bioeconomy models has intensified interest in integrated waste-to-energy systems that simultaneously recover energy, materials, and nutrients from organic biomass streams. In the West African and sub-Saharan context, significant quantities of agricultural and municipal organic waste — including cassava peel, palm empty fruit bunch (EFB), sugarcane bagasse, rice husk, municipal solid waste, and animal manure — represent underutilised feedstock resources with substantial thermodynamic energy content.

Anaerobic Digestion (AD) is an established biological process in which microbial communities decompose organic matter in an oxygen-free environment to produce biogas — a mixture of methane (CH_4) and carbon dioxide (CO_2) — and digestate, a nutrient-rich residue. The methane-rich biogas constitutes a high-value renewable energy carrier for electricity and heat generation via Combined Heat and Power (CHP) units. AD performance is governed by a set of critical process parameters: pH, temperature, organic loading rate (OLR), hydraulic retention time (HRT), total solids (TS) and volatile solids (VS) content, the carbon-to-nitrogen ratio, and the presence of inhibitory compounds such as ammonia and volatile fatty acids (Sarker et al., 2019; Tabatabaei et al., 2018).

Hydrothermal Carbonization (HTC) is a thermochemical process that converts wet biomass and organic residues — including wet digestate from AD — into hydrochar, a stable, carbon-dense solid fuel with properties similar to lignite coal, under subcritical water conditions at temperatures of 180–260°C and autogenous pressures of 10–50 bar (Pauline & Joseph, 2020; Czerwinska et al., 2022). A defining advantage of HTC over conventional pyrolysis or torrefaction is its ability to process feedstocks without prior drying — a significant energy efficiency benefit when treating wet digestate streams. HTC products serve diverse applications including solid biofuel for power generation, soil amendment, carbon sequestration adsorbent, and biogas purification medium.

The integration of AD and HTC into a unified biorefinery delivers multiple synergistic advantages: improved overall energy recovery, elimination of the digestate disposal problem, reduced processing costs, and enhanced material circularity (Cavali et al., 2022; Ipiates et al., 2021). Aragón-Briceño et al. (2021) demonstrated that integrating HTC with AD of sewage sludge can increase net energy production by up to ten times compared to standalone AD, confirming the thermodynamic and economic case for integration.

The reference study by Potrč et al. (2025) advances the field by developing a Mixed-Integer Nonlinear Programming (MINLP) model that simultaneously optimises the AD-HTC process flowsheet variables and the biomass supply chain network across a three-layer superstructure. Applied to Slovenia with four feedstocks — corn silage, cattle manure, sunflower cake, and whey — the model identifies an optimal solution yielding an annual after-tax profit of 23.13 million USD, producing 245.70 GWh/yr of electricity, 298.83 GWh/yr of heat, and 185.08 kt/yr of hydrochar from five plant locations. This report contextualises this methodology within MEG 315 thermodynamics and extends its analysis to a Nigerian feedstock context.

3. Study Objectives

This technical report sets out to achieve the following objectives:

- Contextualise the AD-HTC integrated biorefinery of Potrč et al. (2025) within the applied thermodynamics framework of MEG 315, explicitly linking the system's energy flows to the First and Second Laws of Thermodynamics, gas power cycle theory, vapour power cycle analysis, and heat integration methodology.
- Develop the thermodynamic state-point analysis of the biogas ICE modelled as an ideal Otto cycle, including T-s diagram construction, compression and expansion temperature calculations, net specific work output, and thermal efficiency as a function of compression ratio.
- Perform Rankine steam cycle analysis for the waste heat recovery bottoming cycle at a fixed boiler pressure of 6.0 MPa (60 bar), including h-s Mollier diagram state-point calculations for all four cycle states (pump inlet, pump outlet, turbine inlet, turbine outlet), isentropic turbine efficiency correction, and steam quality determination.

- Apply Pinch Analysis methodology to the system heat integration problem — quantifying the CHP thermal output, AD mesophilic heating demand, and HTC reactor thermal input, and determining the net heat surplus or deficit for a representative feedstock scenario.
- Develop and present a daily mass balance for a representative 10 t/day biomass feed, tracing all streams from feedstock through biogas, digestate, and hydrochar production, with reference to the feedstock database spanning locally relevant West African biomass types.
- Critically evaluate the limitations and engineering gaps in the Potrč et al. (2025) methodology and identify directions for future thermodynamic and technoeconomic research.

4. Methodology

The methodology integrates three analytical components: (i) mass balance modelling of the AD-HTC process flowsheet; (ii) thermodynamic cycle analysis of the gas power (Otto) and vapour power (Rankine) cycles; and (iii) heat integration analysis via Pinch methodology. All calculations are performed for a representative base case using cow manure as the primary feedstock at a feed rate of 10 t/day, with reference values drawn from the MEG 315 dashboard simulation and the steam property tables at $P = 6.0$ MPa.

4.1 AD-HTC Process Flowsheet

The integrated AD-HTC process flowsheet comprises the following unit operations, adapted from Potrč et al. (2025) for the MEG 315 demonstration case:

- Feed Storage (STOR): Biomass feedstock arrives from supply zones and is stored prior to processing. For the base case, 10,000 kg/day of cow manure is assumed (TS = 18%, VS = 80% of TS).
- Mixer 1 (MXR1): All feed streams — fresh feedstock, recycled liquid digestate from the filter press, and recycled HTC process water — are combined. Dry matter content of the combined stream is maintained at 2–15% for feasible AD operation.
- Anaerobic Digestion (AD): Biogas volume is calculated as: $V_{\text{biogas}} = m_{\text{VS}} \times Y_{\text{biogas}}$, where m_{VS} is the volatile solids mass flow (kg/day) and Y_{biogas} is the specific biogas yield ($\text{m}^3/\text{kg VS}$). For cow manure, $Y_{\text{biogas}} = 0.280 \text{ m}^3/\text{kg VS}$ with 60% CH_4 content. At 10 t/day feed: $\text{VS mass} = 10,000 \times 0.18 \times 0.80 = 1,440 \text{ kg/day}$; $V_{\text{biogas}} = 1,440 \times 0.280 = 403.2 \text{ m}^3/\text{day}$.
- Combined Heat and Power (CHP): Biogas fuels a reciprocating gas engine (ICE), modelled thermodynamically as an ideal Otto cycle. Electrical and thermal energy are produced at conversion efficiencies determined by the Otto cycle thermal efficiency and a 45% waste heat recovery factor respectively.

- Filter Press (PRESS): AD digestate is dewatered. The solid fraction (25% dry matter assumed) proceeds to HTC. The liquid fraction is recycled to MXR1. Dry digestate = $VS_{in} \times (1 - 0.60) = 1,440 \times 0.40 = 576 \text{ kg/day}$.
- HTC Reactor: Solid digestate is processed under subcritical water conditions at $T_{HTC} = 180\text{--}260^\circ\text{C}$. Gaseous by-products (primarily CO_2) account for 5–10% of input mass. Hydrochar yield (dry basis) for cow manure = 48%; hydrochar production = $576 \times 0.48 = 277 \text{ kg/day}$ at HHV = 14,500 kJ/kg.
- Filtration (FILT) and Drying (DRY): Hydrochar is dewatered and dried to 80–90% dry matter content and distributed as solid biofuel or soil amendment.

Table 1 summarises the daily mass balance for the base case (cow manure, 10 t/day feed rate):

Stream	Parameter	Formula	Value
Feedstock	Feed rate	Input	10,000 kg/day
Feedstock	Total Solids (TS)	$\dot{m}_{\text{feed}} \times 0.18$	1,800 kg/day
AD input	Volatile Solids (VS)	$TS \times 0.80$	1,440 kg/day
AD output	Biogas volume	$VS \times 0.280 \text{ m}^3/\text{kg VS}$	403.2 m ³ /day
Biogas	Methane volume	$V_{\text{biogas}} \times 0.60$	241.9 m ³ CH ₄ /day
Biogas	Methane LHV energy	$V_{\text{CH}_4} \times 35.8 \text{ MJ/m}^3$	8,660 MJ/day
Digestate	Dry digestate	$VS \times (1 - 0.60)$	576 kg/day
HTC output	Hydrochar mass	$\text{Dry digestate} \times 0.48$	277 kg/day
HTC output	Hydrochar energy (HHV)	$277 \text{ kg} \times 14,500 \text{ kJ/kg}$	4,016 MJ/day

Table 1. Daily mass balance — cow manure base case (10 t/day feed rate).

4.2 Otto Cycle — Biogas ICE Thermodynamic Analysis

The CHP gas engine is modelled as an ideal Otto cycle operating on the air-standard assumption. The Otto cycle consists of four thermodynamic processes between four state points, as shown on the T-s diagram: isentropic compression (1→2), constant-volume heat addition (2→3), isentropic expansion (3→4), and constant-volume heat rejection (4→1).

For the base case ($T_1 = 298 \text{ K}$, compression ratio $r = 10$, peak temperature $T_3 = 1200 \text{ K}$, $\gamma = 1.4$ for air, $\gamma_{\text{gas}} = 1.33$ for combustion products, $c_{p,\text{air}} = 1.005 \text{ kJ/kg}\cdot\text{K}$, $c_{p,\text{gas}} = 1.148 \text{ kJ/kg}\cdot\text{K}$):

Otto Cycle State-Point Calculations (Base Case)

State 2 — Isentropic Compression: $T_2 = T_1 \cdot r^{(\gamma-1)} = 298 \times 10^{0.4} = 748.4 \text{ K}$

State 4 — Isentropic Expansion: $T_4 = T_3 / r^{(\gamma-1)} = 1200 / 10^{0.4} = 477.7 \text{ K}$

Specific compression work: $W_{\text{comp}} = c_{p,\text{air}} \times (T_2 - T_1) = 1.005 \times (748.4 - 298) = 452.6 \text{ kJ/kg}$

Specific expansion work: $W_{\text{turb}} = c_{p,\text{gas}} \times (T_3 - T_4) = 1.148 \times (1200 - 477.7) = 829.5 \text{ kJ/kg}$

Net specific work: $W_{\text{net}} = W_{\text{turb}} - W_{\text{comp}} = 829.5 - 452.6 = 376.9 \text{ kJ/kg}$

Heat input: $Q_{\text{in}} = c_{p,\text{gas}} \times (T_3 - T_2) = 1.148 \times (1200 - 748.4) = 518.7 \text{ kJ/kg}$

Otto cycle thermal efficiency: $\eta_{\text{Otto}} = W_{\text{net}} / Q_{\text{in}} = 376.9 / 518.7 = 0.727 \rightarrow 72.7\%$

Note: Air-standard Otto efficiency formula: $\eta = 1 - 1/r^{(\gamma-1)} = 1 - 1/10^{0.4} = 0.602 \text{ (60.2\%)}$

Net electrical power ($\dot{m}_{\text{air}} = 50 \text{ kg/s}$): $P_{\text{elec}} = \dot{m}_{\text{air}} \times W_{\text{net}} = 50 \times 376.9 = 18,845 \text{ kW}$

Recoverable thermal power (45% recovery): $Q_{\text{therm}} = \dot{m}_{\text{air}} \times c_{p,\text{gas}} \times (T_4 - T_1) \times 0.45$
 $= 50 \times 1.148 \times (477.7 - 298) \times 0.45 = 4,634 \text{ kW}$

State	Process	Temperature (K)	Entropy (kJ/kg·K)	Key Relation
1 — Intake	Start of cycle	298.0	$s_1 = 1.000 \text{ (ref)}$	Ambient conditions
2 — Compression	Isentropic ($s = \text{const}$)	748.4	$s_2 = s_1 = 1.000$	$T_2 = T_1 \cdot r^{(\gamma-1)}$
3 — Combustion	Const. volume (Q_{in} added)	1200.0	$s_3 = s_2 + c_{p,\text{gas}} \cdot \ln(T_3/T_2)$	Peak temperature
4 — Expansion	Isentropic ($s = \text{const}$)	477.7	$s_4 = s_3$	$T_4 = T_3 / r^{(\gamma-1)}$

Table 2. Otto cycle T-s state points — base case ($T_1 = 298 \text{ K}$, $r = 10$, $T_3 = 1200 \text{ K}$).

4.3 Rankine Steam Cycle Analysis

CHP exhaust heat drives a steam bottoming cycle. The cycle is analysed as an ideal Rankine cycle at a fixed boiler pressure of $P_{\text{boiler}} = 6.0 \text{ MPa}$ (60 bar). Steam properties are interpolated from superheated steam tables at 60 bar. The condenser operates at $P_{\text{cond}} = 0.06 \text{ bar}$. Turbine isentropic efficiency $\eta_{\text{st}} = 0.85$; pump efficiency $\eta_{\text{p}} = 0.80$.

The four states of the Rankine cycle are:

- State A (Pump Inlet / Condenser Exit): Saturated liquid at $P_{\text{cond}} = 0.06$ bar. From steam tables: $h_A = 359.9$ kJ/kg, $s_A = 1.145$ kJ/kg·K.
- State B (Pump Outlet / Boiler Inlet): Compressed liquid after isentropic pumping. $w_{\text{pump}} = v_f \times (P_{\text{boiler}} - P_{\text{cond}}) \times 100 = 0.001 \times (60 - 0.06) \times 100 = 5.994$ kJ/kg. Actual: $h_B = h_A + w_{\text{pump}} / \eta_p = 359.9 + 5.994/0.80 = 367.4$ kJ/kg.
- State C (Turbine Inlet): Superheated steam at $P = 6.0$ MPa, $T = 500^\circ\text{C}$. From steam tables: $h_C = 3,423.1$ kJ/kg, $s_C = 6.8826$ kJ/kg·K.
- State D (Turbine Outlet): Expanded steam. Isentropic enthalpy: h_{Ds} determined from quality calculation at condenser pressure. Actual $h_D = h_C - \eta_{st} \times (h_C - h_{Ds})$. Steam quality $x_{\text{act}} = (h_D - h_f) / h_{fg}$.

State	Description	P (bar)	T (°C)	h (kJ/kg)	s (kJ/kg·K)
A — Pump In	Sat. liquid at P_{cond}	0.06	85.9	359.9	1.1450
B — Pump Out	Compressed liquid	60.0	~86.3	367.4	1.1450
C — Turbine In	Superheated steam (500°C)	60.0	500	3,423.1	6.8826
D — Turbine Out	Wet/dry steam ($\eta_{st} = 0.85$)	0.06	85.9	~2,545	~7.5

Table 3. Rankine cycle h-s state points — $P_{\text{boiler}} = 6.0$ MPa, $T_{\text{steam}} = 500^\circ\text{C}$, $P_{\text{cond}} = 0.06$ bar.

Rankine Cycle Performance Metrics

Net specific work: $w_{\text{net}} = (h_C - h_D) - w_{\text{pump}}/\eta_p \approx (3423.1 - 2545) - 7.5 = 870.6$ kJ/kg

Boiler heat input: $q_{\text{boiler}} = h_C - h_B = 3423.1 - 367.4 = 3055.7$ kJ/kg

Rankine cycle efficiency: $\eta_{\text{Rankine}} = w_{\text{net}} / q_{\text{boiler}} = 870.6 / 3055.7 \approx 28.5\%$

4.4 Heat Integration — Pinch Analysis

The heat integration problem for the AD-HTC system involves three streams:

- CHP thermal output (hot stream): Recoverable exhaust heat from the biogas ICE.
- AD heating demand (cold stream 1): Heat required to maintain the anaerobic digester at mesophilic temperature ($T_{\text{AD}} = 37^\circ\text{C}$) from ambient inlet temperature T_{amb} .
- HTC reactor demand (cold stream 2): Heat required to raise wet digestate from $T_{\text{AD}} = 37^\circ\text{C}$ to HTC operating temperature $T_{\text{HTC}} = 180\text{--}260^\circ\text{C}$.

The Pinch Point methodology determines the minimum temperature approach in a heat exchanger network. For the base case (10 t/day cow manure, $T_1 = 298 \text{ K} = 25^\circ\text{C}$, $T_{\text{HTC}} = 220^\circ\text{C}$):

Pinch Analysis — Heat Balance Calculations

$$\begin{aligned} Q_{\text{AD}} &= (\dot{m}_{\text{feed}} / 86,400 \text{ s}) \times c_{p,\text{water}} \times (T_{\text{AD}} - T_{\text{amb}}) \\ &= (10,000 / 86,400) \times 4,180 \times (37 - 25) = 5.81 \text{ kW} \end{aligned}$$

$$\begin{aligned} Q_{\text{HTC}} &= (\dot{m}_{\text{digestate_wet}} / 86,400 \text{ s}) \times c_{p,\text{slurry}} \times (T_{\text{HTC}} - T_{\text{AD}}) \\ &= (9,597 / 86,400) \times 4,000 \times (220 - 37) = 81.6 \text{ kW} \end{aligned}$$

$$\begin{aligned} Q_{\text{avail,CHP}} &= \dot{m}_{\text{air}} \times c_{p,\text{gas}} \times (T_4 - T_1) \times 0.45 \\ &= 50 \times 1.148 \times (477.7 - 298) \times 0.45 = 4,634 \text{ kW} \end{aligned}$$

Heat Balance: $\Delta Q = Q_{\text{avail}} - Q_{\text{AD}} - Q_{\text{HTC}} = 4,634 - 5.81 - 81.6 = +4,546.6 \text{ kW SURPLUS}$

The large heat surplus confirms that CHP exhaust heat greatly exceeds the internal process heating demands for this base case. In practice, this surplus would either drive additional Organic Rankine Cycle (ORC) generation from low-grade waste heat, supply district heating networks, or be rejected via cooling towers if no additional recovery is implemented.

5. Critical Evaluation

5.1 Limitations and Engineering Gaps

While the methodology of Potrč et al. (2025) represents a significant advance in the simultaneous optimisation of AD-HTC process and supply chain, several important engineering limitations warrant critical attention:

- **Simplified biogas yield model:** The biogas conversion factors employed are static empirical constants (m^3 biogas per tonne of feedstock). A mechanistic model such as the IWA Anaerobic Digestion Model No. 1 (ADM1) would capture the dynamic effects of substrate inhibition, pH fluctuations, temperature transients, and microbial community composition shifts with significantly greater accuracy — but at substantially higher computational cost and parametric data requirements.
- **Fixed HTC conversion parameters:** HTC yield and hydrochar quality are modelled using fixed bounds on gaseous product fraction (5–10%) and filtration dry matter recovery (70–80%). In practice, HTC operating temperature and residence time are the primary

determinants of hydrochar yield, calorific value, surface area, and functional group chemistry — none of which are captured by the constant-conversion-factor approach.

- **Otto cycle idealisation:** The air-standard Otto cycle assumption ignores real-gas effects, combustion kinetics, heat transfer losses through cylinder walls, mechanical friction, and the variable specific heat of combustion products across the temperature range of interest. Real biogas ICE thermal efficiencies are typically 30–38%, compared to the air-standard Otto prediction of ~60% at $r = 10$ — a significant overestimation that would cause the heat integration analysis to overstate available CHP thermal output.
- **No Second Law (exergy) analysis:** The system energy balance accounts for First Law quantities only. A complete exergy analysis would quantify the irreversibilities concentrated in the combustion process (the largest exergy destruction site in any combustion-driven cycle), in heat transfer across finite temperature differences in the AD and HTC heat exchangers, and in mixing of streams at different temperatures and compositions — revealing where thermodynamic losses are concentrated and guiding targeted efficiency improvements.
- **No Life Cycle Assessment (LCA):** Environmental performance claims — particularly regarding greenhouse gas (GHG) emission reductions and circular bioeconomy contributions — cannot be substantiated without a full LCA. The authors acknowledge this gap and identify it as future work, including the need to incorporate carbon, nitrogen, phosphorus, heavy metal, and other substance balances in digestate and hydrochar streams.
- **Geographic and feedstock specificity:** The Potrč et al. (2025) model parameters — energy prices based on Slovenian day-ahead electricity market data, feedstock costs, and transportation infrastructure — are calibrated for Slovenia. Direct application to the Nigerian context would require full recalibration of all economic inputs, feedstock characterisation data, and supply chain logistics parameters. West African feedstocks such as cassava peel, palm EFB, sugarcane bagasse, and rice husk have distinct TS/VS profiles, biogas yields, and HTC hydrochar properties that differ from European silage and manure feedstocks.
- **HTC process water inhibition:** HTC process water contains high concentrations of furfurals, phenolics, and other compounds with high chemical oxygen demand (COD) that can significantly inhibit AD microbial activity when recirculated to the digester. The constant conversion factor approach in the model does not capture this inhibition effect, which may substantially reduce real biogas yields compared to model predictions.

6. Relevance to MEG 315 — Applied Thermodynamics

The AD-HTC integrated biorefinery system connects directly to every core area of MEG 315 at the University of Lagos. The following analysis demonstrates these linkages explicitly:

6.1 First and Second Laws of Thermodynamics

The CHP unit is governed by the First Law: chemical energy stored in biogas (quantified by the Lower Heating Value of methane, $\text{LHV}_{\text{CH}_4} = 35.8 \text{ MJ/m}^3$) is converted to useful electrical output, useful thermal output, and unavoidable losses. The overall CHP First Law efficiency $\eta_{\text{CHP}} = (W_{\text{elec}} + Q_{\text{therm}}) / Q_{\text{biogas_in}}$. For the air-standard Otto cycle base case, this represents the total energy conversion fraction for the gas engine.

A complete Second Law (exergy) analysis would further quantify irreversibilities concentrated in the combustion process — which, due to the large temperature difference between the flame ($\sim 2,000 \text{ K}$) and the working fluid, is the dominant source of exergy destruction in any combustion cycle. Heat transfer in the AD reactor (at $\sim 37^\circ\text{C}$) and in the HTC reactor (at $180\text{--}260^\circ\text{C}$) across finite temperature differences also generates entropy and destroys exergy. The Second Law efficiency $\eta_{\text{II}} = W_{\text{actual}} / W_{\text{reversible}}$ reveals how far the system operates from its thermodynamic ideal.

6.2 Gas Power Cycles — Otto Cycle

The biogas-fuelled CHP unit employs a reciprocating internal combustion engine (ICE). For small-to-medium scale AD installations — the scale relevant to the Potrč et al. demonstration case and to the Nigerian biomass context — the appropriate thermodynamic cycle model is the ideal Otto cycle (not the Brayton cycle, which applies to open-cycle gas turbines). The Otto cycle's four processes — isentropic compression ($1 \rightarrow 2$), constant-volume heat addition ($2 \rightarrow 3$), isentropic expansion ($3 \rightarrow 4$), and constant-volume heat rejection ($4 \rightarrow 1$) — directly describe the work output and thermal efficiency of the biogas CHP engine.

The Otto cycle thermal efficiency $\eta = 1 - 1/r^{(\gamma-1)}$ shows that efficiency increases monotonically with compression ratio r . For biogas engines, practical compression ratios of 8–12 give air-standard efficiencies of 56–63%, with real efficiencies of 30–38% after accounting for combustion irreversibilities, heat losses, and mechanical friction. The T-s diagram for the Otto cycle provides direct visual representation of the heat added, heat rejected, and net work area enclosed by the cycle.

6.3 Vapour Power Cycles — Rankine Cycle

Where CHP thermal output drives steam generation for a bottoming-cycle, Rankine cycle analysis applies. The MEG 315 analysis fixes the boiler pressure at 6.0 MPa (60 bar) — a choice that reflects industrial steam generation practice and places the superheated steam properties in a well-defined region of the Mollier h-s diagram. State-point calculations on the Mollier diagram — pump work ($A \rightarrow B$), boiler heat addition ($B \rightarrow C$), turbine work extraction ($C \rightarrow D$), and condenser heat rejection ($D \rightarrow A$) — constitute the core computational exercise.

At the sub-megawatt thermal scale typical of small AD installations, Organic Rankine Cycle (ORC) systems using low-boiling-point working fluids such as R245fa, isopentane, or toluene can recover low-grade CHP waste heat more effectively than conventional steam Rankine cycles, operating

at lower boiler temperatures and pressures that better match the available waste heat temperature level.

6.4 Heat Integration and Pinch Analysis

The heat balance in the AD-HTC system — where CHP waste heat must simultaneously satisfy the AD mesophilic heating demand and the HTC reactor thermal input — is a direct application of Pinch Analysis methodology. The Pinch Point defines the minimum temperature approach between hot and cold streams in the heat exchanger network; below this temperature difference, heat transfer becomes thermodynamically infeasible without additional heating utility input.

Optimising the heat exchanger network to maximise internal heat recovery reduces external energy input and operating cost. This is precisely the engineering trade-off that the Potrč et al. (2025) MINLP model resolves by constraining heat flows between the CHP, AD, and HTC units within the process superstructure. Correctly identifying the Pinch temperature and the minimum hot utility (MHU) and minimum cold utility (MCU) requirements using Composite Curve and Grand Composite Curve analysis is the systematic approach to this problem.

6.5 Systems-Level Engineering Design

Perhaps the most important MEG 315 lesson embedded in the AD-HTC biorefinery is that thermodynamic engineering design cannot be reduced to optimising individual unit operations in isolation. The simultaneous optimisation of process flowsheet variables and supply chain logistics in the Potrč et al. (2025) framework demonstrates that system-level interactions — heat integration between CHP, AD, and HTC units; feedstock routing across a three-layer supply network; product distribution to multiple demand locations — determine the overall thermodynamic and economic performance of the biorefinery far more than any individual unit's local efficiency.

This systems perspective — identifying interactions, constraints, and trade-offs across the entire process boundary — is the hallmark of fourth-year engineering systems thinking and the central intellectual challenge of process integration in MEG 315.

7. Conclusion

This report has contextualised the AD-HTC integrated biorefinery methodology of Potrč et al. (2025) within the thermodynamic framework of MEG 315 Applied Thermodynamics at the University of Lagos, extending the analysis to a Nigerian feedstock context and developing explicit thermodynamic calculations for the system's key cycles and heat integration challenge.

The biogas ICE modelled as an Otto cycle yields a thermal efficiency of approximately 60.2% (air-standard) at compression ratio $r = 10$, with real efficiencies of 30–38% in practice. The Rankine steam bottoming cycle at $P = 6.0$ MPa and $T_{\text{steam}} = 500^{\circ}\text{C}$ achieves a cycle efficiency of approximately 28.5% ($\eta_{\text{st}} = 0.85$). Pinch analysis confirms a substantial CHP heat surplus of

over 4,500 kW for the base case — confirming that the internal AD and HTC heating demands are comfortably satisfied by the available waste heat, with significant potential for additional ORC generation or district heating supply.

The daily mass balance for 10 t/day cow manure feed yields 403.2 m³/day biogas, 277 kg/day hydrochar (HHV = 14,500 kJ/kg, energy content 4,016 MJ/day), confirming the dual-fuel nature of the integrated system. The system's zero-waste design — where AD digestate is fully valorised through HTC rather than disposed — represents both an economic advantage and a circular economy contribution.

Critically, the Potrč et al. (2025) methodology carries important limitations: the air-standard Otto cycle significantly overestimates real ICE efficiency; fixed HTC conversion parameters do not capture the temperature and residence time dependence of hydrochar yield and quality; and the absence of an exergy analysis and LCA leaves thermodynamic and environmental performance incompletely characterised. Future work should incorporate Second Law exergy analysis, ADM1-based mechanistic AD modelling, ORC cycle optimisation for low-grade heat recovery, and technoeconomic analysis calibrated to West African feedstock and energy market conditions.

The integrated AD-HTC biorefinery demonstrates that thermodynamic optimisation at the system level — encompassing cycle selection, heat integration, mass balance, and supply chain logistics — is inseparable from economic optimisation. This is the defining challenge of advanced process engineering, and the central lesson of MEG 315 at 400 level.

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